Abitsync Mini-Tasks - PART A

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- *This mini-task gas been broken into 2 seperate parts. Part B will be released seperately.*
- *Feel free to skip to definition 2.1 if you are confident with basic probability.*

§1 Hello again, my old friend

Yep. Probability's back! But before you decide this project isn't for you, it's important that you understand that this version of probability is quite different from the vague probability you have learnt earlier and that you might just learn to love it by the end of this mini-project.

To recap, here are few obvious things about probability of different events in an experiment:

1. Probability can never be negative.
2. The sum of the probabilities of all atomic events of an experiment must be 1.

To kick things off, here are a few simple probability questions for you [no need to submit :)]

Problems

1. If two balanced dice are rolled, what is the probability that the sum of the two numbers that appear will be odd? 1/2
2. If three fair coins are tossed, what is the probability that all three faces will be the same? 1/4
3. Consider an experiment in which a fair coin is tossed once and a balanced die is rolled once. What is the probability that a head will be obtained on the coin and an odd number will be obtained on the die? 1/4

Definition 1.1. **Sample space** S: This is the set of all possible outcomes of an experiment. Note that each outcome has a specific probability associated with it.

Try finding the sample spaces and the associated probabilities for the above problems.

§2 Variables and randomness?

Alright. Now that you hopefully remember a bit a probability let's get down to business.

Some of you might have vague notions of what are called random variables (referred to in some places as RVs). These random variables are at the heart of both our project and much of probability theory.

Definition 2.1. A **Random Variable** X is a map from a sample space of a particular experiment to the set of real numbers i.e. $\omega \in S$ corresponds to a real value X and $X(\omega) = x$

Eg. Consider a coin toss. Then, $S = \{Heads,\ Tails\}$. A few possible random variables which can be defined are:

- $X(Heads) = 1,\ X(Tails) = 0$

- $X(Heads) = 10^7,\ X(Tails) = -10$
- $X(Heads) = X(Tails) = 0$

Note that multiple ω might map to the same real value x. Thus the random variable function need not be one-one or injective.

Definition 2.2. The **probability mass function** p_X of a random variable X is defined as $P(X = x)$ i.e. $p_X(x)$ is that probability that X takes on the value x. This is obviously just the sum of the probabilities of all events ω (or objects) in sample space S such that $X(w) = x$.

$$p_X(x) := P(\{\omega \in S \mid X(\omega) = x\})$$

One might wonder what the use of such seemingly useless definitions are. It seems like we lose information about the sample space by calculating a RV (due to the non-bijective nature).

However, an important point to note is that analysis of the sample space itself (or rather, the probability of every event) might be impossible in certain situations. What sort of situations?

Eg. *Imagine you threw a dart at a wall from far away. Realistically, you cannot predict the probability of it landing at each point in the wall due to the nature of external factors such as wind, weight of the dart etc. But you know for a fact that the dart is on the wall (assume a somewhat competent thrower).*

Thus a random variable (defined as being 1 if the wall is hit and 0 if it isn't), along with its PMF, is completely known even though the sample space itself isn't.

Notice that we can construct a new random variable by scaling or adding 2 existing random variables.

Eg. *Consider the first coin toss example as the random variable X. Then we can define the random variable $Y = 5X$. In this case, getting heads corresponds to an output of 5 which implies a 50 percent chance of getting 5 and a 50 percent chance of getting 0.*

Let us try extending this notion to 2 variables:

Definition 2.3. The *joint PMF* of two random variables X, Y is defined as the probability that the pair of random variables (not necessarily independent) (X, Y) takes the values (x, y) respectively, i.e:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

This is pretty intuitive. This statement simply asserts that given an experiment with two results, the PMF of the experiment is defined as the probability that the result tuple is the same as a given tuple.

I bet that was a lot of text. Now for the fun part.

Problems

- Not negative $P_X(x) \geq 0 \ \forall x$
- Sum = 1 $\sum_x P_X(x) = 1$

1. Can you find some fundamental properties of the PMF which relates to the addition of all the values? Try and prove them. (Hint: It follows from the basic definitions of probability)

   Total probability of all mutually exclusive events in the sample space is 1

2. Which of the following is(are) probability mass functions?

   a) $P(X = x) = \frac{3}{4} \times \frac{1}{4^x}\ ,\ x \in \mathbb{N} \cup \{0\}$

   b) $P(X = x) = \frac{1}{8} \times (x + 1)\ ,\ x \in \mathbb{N} \cup \{0\}$

c) $P(X = x) = \frac{1}{2} \times \frac{1}{2^x}\ ,\ x \in \mathbb{N} \cup \{0\}$

d) $P(X = x) = \frac{1}{4} \times (x+1)\ ,\ x \in \mathbb{N} \cup \{0\}$

X = 0,1,2,3,4,5

3. Suppose that two balanced dice are rolled, and let X denote the absolute value of the difference between the two numbers that appear. Determine and sketch the PMF of X.

P(X=0) = 1/6 P(X=1) = 5/18 P(X=2) = 2/9 P(X=3) = 1/6 P(X=4) = 1/9 P(X=5) = 1/18

4. In the above question, assume the values of the dies correspond to the RVs Y and Z. Find the joint PMF of Y and Z

Y,Z → independent ⇒ P(Y=y, Z=z) = P(Y=y) · P(Z=z)

= 1/6 · 1/6 = 1/36

§3 Continuous probability?

We have looked how probability behaves when the sample space is discrete (the elements can be listed out). However this is obviously not always the case.

Eg. *Consider an antenna receiving some sort of an analog signal with a 5V peak. Obviously the measured voltage at the ends of the antenna can lie anywhere between 0 to 5V. One might ask what the probability of reading a voltage between say, 1V and 2V is.*

In such cases we are forced to deal with what are known as **continuous probability distributions**.

An interesting point to note is that the definition of random variables does not change even while dealing with continuous distributions. It may not be possible to assign a real value by hand to every single event (since there are now infinite events), but it is possible to come up with some sort of continuous mapping.

Consider the previous example. I can assign a random variable X which returns twice the measured numerical value of voltage. Note that this is still a map from the sample space to real numbers.

A consequence of this is that continuous probability distributions can give rise to both discrete and continuous random variables. However, a discrete probability distribution can only give rise to a discrete random variable.

Definition 3.1. The **Cumulative Probability Function** (CDF) $F_X(x)$ of a random variable X is defined as $P(X \leq x)$ i.e. the probability that the result of a random experiment is lesser than a given value x.

Note that this definition holds for both continuous and discrete random variables. Also note that eventually the CDF must tend towards 1 as x tends towards ∞.

One might ask whether there is any analog to PMF for continuous RVs. The answer to that is yes!

Definition 3.2. The **probability distribution function** (PDF) f_X of a random variable X is defined such that:

$$\int_y^{y+\Delta y} f_X(x)\ dx \ = \ P(y \leq x \leq y + \Delta y)$$

This may seem like insufficient information to completely define the PDF but it turns out it actually is. In most cases a much more useful way to find the PDF is simple to differentiate the CDF.

Problem time!

Problems

$F(0) = 0.1 \quad F(1) = 0.4 \quad F(2) = 0.8 \quad F(4) = 1$

1. Suppose that a random variable X can take only the values 2, 0, 1, and 4, and that the probabilities of these values are as follows: $Pr(X = 2) = 0.4$, $Pr(X = 0) = 0.1$, $Pr(X = 1) = 0.3$, and $Pr(X = 4) = 0.2$. Sketch the CDF of X.

2. Find a relation between the CDF and the PDF of a continuous differentiable random variable. Express this both ways, i.e. express CDF in terms of the PDF and also the PDF in terms of the CDF. $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$; $f_X(x) = \frac{d}{dx} F_X(x)$ (CDF, PDF)

3. Classify the following as valid PDFs (given in terms of variable x):

   a) e^x → not possible

   b) e^{-x} → domain : $x \geq 0$

   c) e^{-10x} → $10e^{-10x}$; $x \geq 0$

   d) ${}^{100}C_x\ 0.11^x\ 0.89^{100-x}$ ⇒ $CDF = \sum_{k=0}^{\lfloor x \rfloor} {}^{100}C_k\ (0.11)^k\ (0.89)^{100-k}$

   If they are valid, find their CDFs. If they are not, is it possible to make them valid somehow?

4. Suppose that a point in the xy-plane is chosen at random from the interior of a circle for which the equation is $x^2 + y^2 = 1$; and suppose that the probability that the point will belong to each region inside the circle is proportional to the area of that region. Let Z denote a random variable representing the distance from the center of the circle to the point. Find and sketch the CDF of Z.

$F_{XY}(-\infty, y) = 0$

$F_{XY}(x, -\infty) = 0$

$F_{XY}(\infty, \infty) = 1$

$F_{XY}(x, y)$ → non decreasing in both x and y

rel^n: $f_{XY}(x, y) = \frac{\partial^2 F}{\partial x \partial y}\left(F_{XY}(x, y)\right)$

5. We have seen CDF for a single RV. Can this be extended for a bivariate (a distribution with 2 RVs as seen in the joint PMF section)? If so, provide an expression. Now try and extend it for continuous bivariate RVs and find the corresponding PDF expression.

discrete random variables: $F_{XY}(x, y) = P(X \leq x, Y \leq y) = \sum\sum P(X = i, Y = j)$ $\begin{pmatrix} i \leq x \\ j \leq y \end{pmatrix}$

continuous random variables: $F_{XY}(x, y) = \int_{-\infty}^{t}\int_{-\infty}^{s} f_{XY}(s, t)\,ds\,dt$

§4 Conditional Probability

Let there be two discrete random variables X and Y. Let us assume at the moment we are studying Y individually. Does the probability of Y taking an arbitrary value y change depending on the value X takes. The answer is yes!! (unless X and Y are independent).

Definition 4.1. *Revising Basics:* The probability of occurence of an event B conditioned on another event A is denoted by $P(B/A)$

Notice how the probability distribution (or PMF) of the variable Y also changes depending on the value of X.

Definition 4.2. The **Conditional PMF of Y over X = x** denoted by $p_{Y|X=x}$ is given by:

$$p_{Y|X=x}(y) = P(Y = y|X = x)$$

Note that the conditional PMF is a function defined on one single variable Y, not both X and Y.

Similarly for continuous variables:

Definition 4.3. The **Conditional CDF** *of Y over* $X = x$ denoted by $F_{Y|X=x}$ is given by:

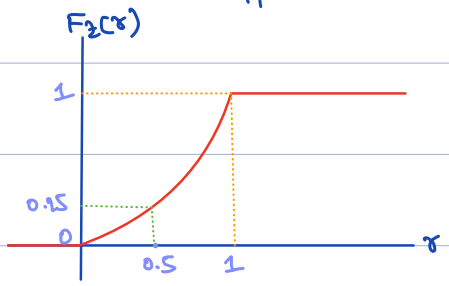
$$F_{Y|X=x}(y) = P(Y \leq y|x \leq X \leq x + \Delta x)$$

Suppose that a point in the xy-plane is chosen at random from the interior of a circle for which the equation is $x^2 + y^2 = 1$; and suppose that the probability that the point will belong to each region inside the circle is proportional to the area of that region. Let Z denote a random variable representing the distance from the center of the circle to the point. Find and sketch the CDF of Z.

$Z = \sqrt{x^2+y^2}$

$F_Z(z) = P(Z \le z) = \frac{\pi z^2}{\pi} = z^2 \quad z \in [0,1]$

CDF:



1. There are 4 blue balls and 6 red balls in a box. Balls are blindly removed one by one. Find the probability of the 3rd ball drawn being red. The answer may feel obvious but we cannot forget that this probability depends on the previous two draws. Hence try to arrive at the answer using only conditional arguments.

$P(\text{red, blue}) = \frac{6}{10} \times \frac{4}{9}$ $\quad P(\text{blue, red}) = \frac{4}{10} \times \frac{6}{9}$

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$P(3^{rd} \text{ red}) = \frac{6}{10} \times \frac{4}{9} \times \frac{5}{8} + \frac{4}{10} \times \frac{6}{9} \times \frac{5}{8} + \frac{4}{10} \times \frac{3}{9} \times \frac{6}{8} + \frac{6}{10} \times \frac{5}{9} \times \frac{4}{8}$

$= \frac{120+120+72+120}{10 \times 9 \times 8} = \frac{432}{720} = \frac{3}{5}$

2. **Monty Hall Problem:** Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a cash prize of 10000 Rs; behind the others, nothing. You pick a door, say No. 1, and the host (who knows what's behind the doors) opens one of the other doors, say No. 3, which has nothing behind it. He then says to you, "Do you want to pick door No. 2?" Should you or should you not switch?

initially,

$P(D1) = 1/3$ $\quad P(D2 \text{ or } D3) = 2/3$

Since there's nothing behind D3; P(D1) still

remains 1/3 (cuz host could have opened either D2 or D3 if prize was in D1) and $P(D_2)$ becomes 2/3. So it would be better to switch to door 2.

3. Suppose that X and Y have a joint continuous distribution with joint PDF $f_{XY}(x,y) = e^{-x} when 0 \le y < x < \infty$ and 0 elsewhere
   - Find the conditional CDF $F_{Y/X=k}$ where k is a constant and $0 < k < \infty$.
   - Use this to find the conditional PDF.

$$F_{Y|X=k}(y) = P(Y \le y \mid X = k) = \int_{-\infty}^{y} \frac{f_{XY}(k,y)\,dy}{f_X(k)}$$

$$f_{XY}(k,y) = \begin{cases} e^{-k} & y \in [0,k) \\ 0 & \text{else} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy = \int_0^x e^{-x}\,dy = xe^{-x}$$

for $0 \le y < k$:

$$F_{Y|X=k}(y) = \frac{\int_0^y e^{-k}\,dy}{ke^{-k}} = y/k$$

for $y \ge k$: $F_{Y|X=k}(y) = 1$

for $y < 0$: $F_{Y|X=k}(y) = 0$

$$f_{Y|X=k}(y) = \frac{d}{dy} F_{Y|X=k}(y) = \begin{cases} 0 & y < 0 \\ 1/k & y \in [0,k) \\ 0 & y \ge k \end{cases}$$

Definition 4.4. The **Conditional PDF** *of* Y *over* $X = x$ denoted by $f_{Y|X=x}$ is the derivative of the Conditional CDF at that point:

$$\int_{y}^{y+\Delta y} f_{Y|X=x}(y) = P(y \leq Y \leq y + \Delta y | x \leq X \leq x + \Delta x)$$

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